

M:\PROJECTS\F5\MN\MN ERFO FS SPER1909 2022-11\BridgeMicrostation\Design Files\R07_MN ERFO FS SPER1909 2022-11\Reinforcing List - LDGN 12/10/2024

PROJECT	SHEET NUMBER
MN ERFO FS SPER1909 2022-1(1)	R07

REINFORCING STEEL SCHEDULE							DIMENSION TABLE										
Description: SOUTH HEADWALL AND WINGWALLS																	
Itm	Qty	Dia	Length	Mark	Shape	lbs	A	B	C	D	E	F/R	G	H	J	K	O
1	10	5	13-03	#5W1	17	138.20		12-05	0-10								
2	50	5	3-08	#5W2	17	191.22		2-10	0-10								
3	68	5	Varies 13-02 to 4-11	#5W3	17	641.27		Varies 12-04 to 4-01	0-10								
4	10	6	16-03	#6W4	17	244.08		13-09	2-06								
5	50	6	6-09	#6W5	17	506.93		4-03	2-06								
6	68	6	Varies 17-03 to 7-10	#6W6	17	1,229.89		Varies 13-08 to 5-05	2-06								
7	100	5	Varies 4-09 to 0-07	#5W7	2	195.04		Varies 4-09 to 0-07									
8	8	5	29-08	#5W8	2	247.54		29-08									
9	36	5	Varies 10-10 to 1-11	#5W9	2	153.32		Varies 10-10 to 1-11									
10	16	5	20-08	#5W10	12	344.89			18-02	2-06				1-09		1-09	19-11
11	32	5	Varies 18-10 to 4-10	#5W11	12	394.95			Varies 16-04 to 2-04	2-06				1-09		1-09	Varies 18-01 to 4-01
12	4	5	21-06	#5W12	12	89.70			19-00	2-06				Field bend		Field bend	Field bend
13	40	8	Varies 11-02 to 0-04	#8F1	2	614.10		Varies 11-02 to 0-04									
14	104	8	11-09	#8F2	2	3,262.74		11-09									
15	28	5	Varies 35-11 to 26-02	#5F3	2	906.54		Varies 35-11 to 26-02									
16	40	8	Varies 11-01 to 0-08	#8F4	2	627.45		Varies 11-01 to 0-08									
17	120	8	Varies 11-02 to 9-09	#8F5	2	3,350.85		Varies 11-02 to 9-09									
18	48	5	Varies 21-10 to 17-09	#5F6	12	990.85			Varies 19-04 to 15-03	2-06				1-09		1-09	Varies 21-01 to 17-00
19	4	5	11-04	#5F7	12	47.28			8-10	2-06				1-09		1-09	10-07
20	4	5	22-09	#5F8	12	94.91			20-03	2-06				1-11		1-07	21-10
Total	840					14,271.73											

STRUCTURE NO.			STRUCTURE NAME	FEATURES INTERSECTED			LOCATION			U.S. DEPARTMENT OF TRANSPORTATION FEDERAL HIGHWAY ADMINISTRATION OFFICE OF FEDERAL LANDS HIGHWAY	
			KIMBALL CREEK ROAD (RTE FR-304) CULVERT	KIMBALL CREEK			SUPERIOR NATIONAL FOREST				
NO.	DATE	BY	REVISIONS	DESIGNED BY	DRAWN BY	CHECKED BY	PROJECT TEAM LEAD	SCALE	BRIDGE PLAN SHEET	DATE	RG / BRP NO.
				TP	TP	DJL	Christopher Negley	Not to scale	7 of 10	December 2024	BRP-1358

REINFORCING LIST - 1

M:\PROJECTS\F\SMN\MN ERFO FS SPER1909 2022-11\Bridge\Microstation\Bridge Design Files\R08.MN ERFO FS SPER1909 2022-11\Reinforcing List - 2.DGN

12/10/2024

PROJECT	SHEET NUMBER
MN ERFO FS SPER1909 2022-1(1)	R08

REINFORCING STEEL SCHEDULE							DIMENSION TABLE										
Description: NORTH HEADWALL																	
Itm	Qty	Dia	Length	Mark	Shape	lbs	A	B	C	D	E	F/R	G	H	J	K	O
21	40	5	13-03	#5W13	17	552.79		12-05	0-10								
22	50	5	3-08	#5W14	17	191.22		2-10	0-10								
23	40	6	16-03	#6W15	17	976.30		13-09	2-06								
24	50	6	6-09	#6W16	17	506.93		4-03	2-06								
25	100	5	Varies 4-09 to 0-07	#5W17	2	195.04		Varies 4-09 to 0-07									
26	8	5	44-08	#5W18	2	372.70		44-08									
27	36	5	Varies 18-04 to 9-05	#5W19	2	437.71		Varies 18-04 to 9-05									
28	180	8	11-09	#8F9	2	5,647.05		11-09									
29	28	5	44-08	#5F10	2	1,304.45		44-08									
Total	532						10,184.18										

REINFORCING STEEL SCHEDULE							DIMENSION TABLE										
Description: MISC																	
Itm	Qty	Dia	Length	Mark	Shape	lbs	A	B	C	D	E	F/R	G	H	J	K	O
30	222	5	7-06	#5K1	S4	1,736.60	0-10	2-05	1-00	2-05			0-10				
31	6	5	29-10	#5K2	2	186.70		29-10									
32	12	5	19-06	#5K3	12	244.06			17-00	2-06				1-09		1-09	18-09
33	6	5	44-08	#5K4	2	279.52		44-08									
34	8	5	18-00	#5C1	23	150.19			5-06	8-00	4-06			2-09		1-02	
35	8	5	5-00	#5C2	2	41.72		5-00									
Total	262						2,638.79										

GRAND TOTAL: 27,095 lbs

STRUCTURE NO.			STRUCTURE NAME	FEATURES INTERSECTED			LOCATION			U.S. DEPARTMENT OF TRANSPORTATION FEDERAL HIGHWAY ADMINISTRATION OFFICE OF FEDERAL LANDS HIGHWAY	
			KIMBALL CREEK ROAD (RTE FR-304) CULVERT	KIMBALL CREEK			SUPERIOR NATIONAL FOREST				
NO.	DATE	BY	REVISIONS	DESIGNED BY	DRAWN BY	CHECKED BY	PROJECT TEAM LEAD	SCALE	BRIDGE PLAN SHEET	DATE	RG / BRP NO.
				TP	TP	DJL	Christopher Negley	Not to scale	8 of 10	December 2024	BRP-1358

REINFORCING LIST - 2

M:\PROJECTS\F5\NMN\ERFO FS SPER1909 2022-11\BridgeMicrostation\Design Files\R09_MN ERFO FS SPER1909 2022-11\Typical bar bends - 1.DGN

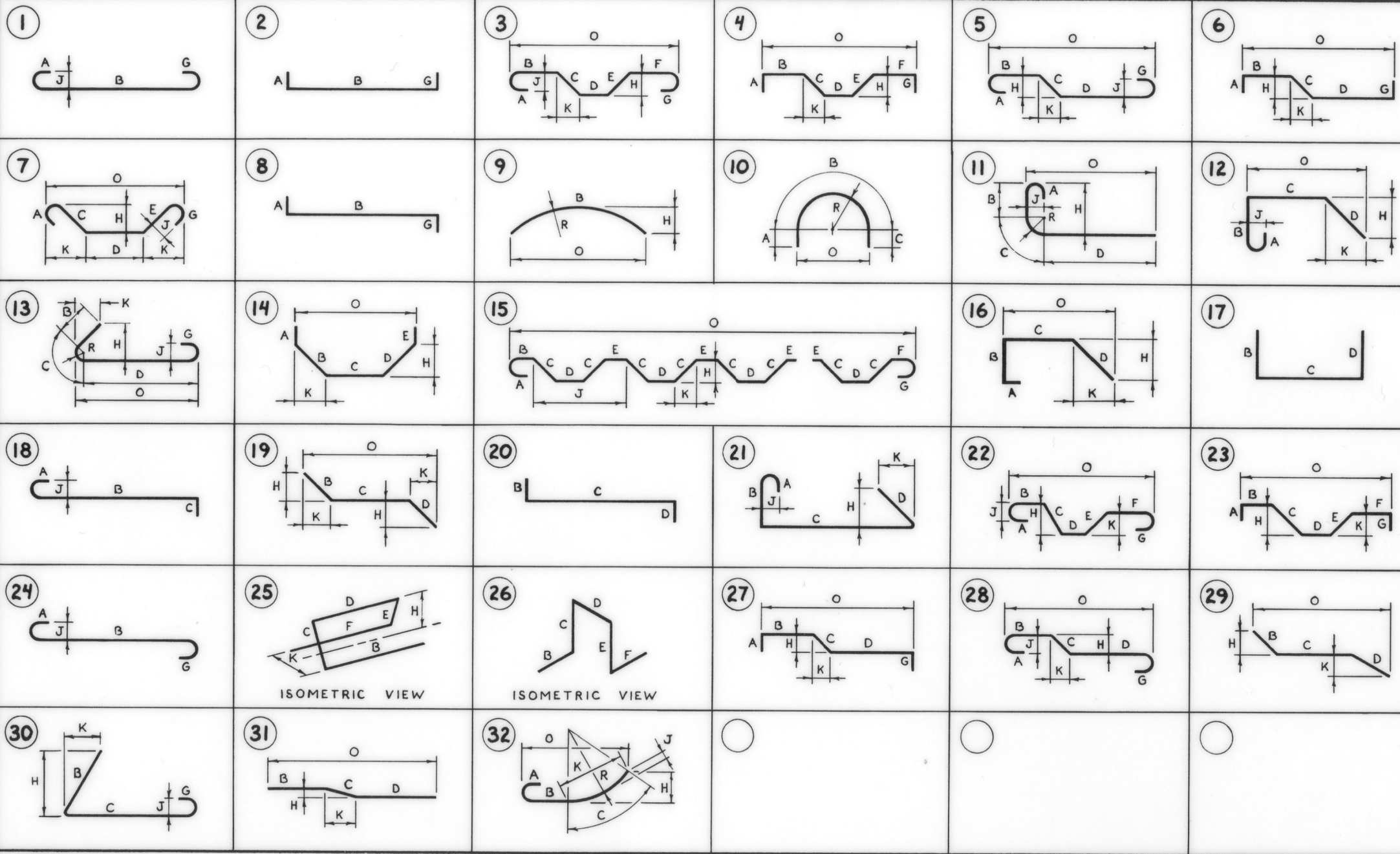
12/10/2024

Applied Systems Associates, Inc.

aSa®/REBAR SYSTEM
STANDARD BAR BENDS

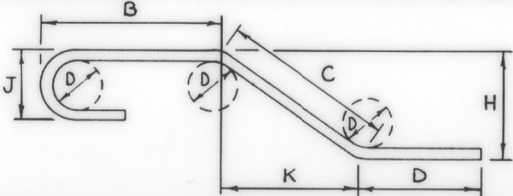
5270 LOGAN FERRY ROAD
MURRYSVILLE, PA 15668
PHONE (724) 733-8700

PROJECT	SHEET NUMBER
MN ERFO FS SPER1909 2022-1(1)	R09



NOTES:

1. aSa Standard Bar Bends include only Types 1-32, T1-T16, and S1-S15.
2. All dimensions are out-to-out of bar except "A" and "G" on standard 180° and 135° hooks.
3. "J" dimension on 180° hooks to be shown only where necessary to restrict hook size, otherwise standard hooks are to be used.
4. Where "J" is not shown, "J" will be kept equal to or less than "H" on truss bars. Where "J" can exceed "H", it should be shown.
5. "H" dimension stirrups to be shown where necessary to fit within concrete.



6. Unless otherwise noted, DIAMETER "D" is the same for all bends and hooks on a bar.
7. Where slope differs from 45° dimensions, "H" and "K" must be shown.
8. Where bars are to be bent more accurately than standard bending tolerances, bending dimensions which require closer fabrication should have limits indicated.
9. Figures in circles show types.
10. For recommended diameter "D", of bends, hooks, etc., see CRSI or ACI tables.
11. Type S1-S15, T1-T16 apply to bar sizes #3 through #8.
12. "J" dimension on type T14, T16 is assumed to be equal to "K" if not specified.

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NO.	DATE	BY	REVISIONS	DESIGNED BY	DRAWN BY	CHECKED BY	PROJECT TEAM LEAD	SCALE	BRIDGE PLAN SHEET	DATE	RG / BRP NO.
				TP	TP	DJL	Christopher Negley	Not to scale	9 of 10	December 2024	BRP-1358

TYPICAL BAR BENDS - 1

M:\PROJECTS\F\SMN\MN ERFO FS SPER1909 2022-11\BridgMicrostation\Bridg Design Files\10_MN ERFO FS SPER1909 2022-11\Typical bar bends - 2.DGN

12/10/2024

Applied Systems Associates, Inc.										aSa®/REBAR SYSTEM STANDARD BAR BENDS		5270 LOGAN FERRY ROAD MURRYSVILLE, PA 15668 PHONE (724) 733-8700																																									
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SPIRAL NOTES: J = TURNS AT 'F' SPACING. K = EXTRA TURNS (HALF T. & B.). (XL) PLAIN SPIRAL WITH SPACERS LOOSE. (XM) PLAIN SPIRAL WITH SPACERS MOUNTED.																																																					
NOTES: - aSa Standard Bar Bends include only Types 1-32, T1-T16, and S1-S15. - All dimensions are out-to-out of bar except "A" and "G" on standard 180° and 135° hooks. "J" dimension on 180° hooks to be shown only where necessary to restrict hook size, otherwise standard hooks are to be used. - Where "J" is not shown, "J" will be kept equal to or less than "H" on truss bars. Where "J" can exceed "H", it should be shown. "H" dimension stirrups to be shown where necessary to fit within concrete. - Unless otherwise noted, DIAMETER "D" is the same for all bends and hooks on a bar. - Where slope differs from 45° dimensions, "H" and "K" must be shown. - Where bars are to be bent more accurately than standard bending tolerances, bending dimensions which require closer fabrication should have limits indicated. - Figures in circles show types. - For recommended diameter "D", of bends, hooks, etc., see CRSI or ACI tables. - Type S1-S15, T1-T16 apply to bar sizes #3 through #8. "J" dimension on type T14, T16 is assumed to be equal to "K" if not specified. 5-88																																																					
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TYPICAL BAR BENDS - 2